SQL Master Worksheet (Updated)

Includes schema definitions and practice questions. Blank pages removed. CASE statements and CTEs are intentionally excluded.

Database Schemas

Customers

```
CREATE TABLE customers (
 customer_id INT PRIMARY KEY,
 customer_name VARCHAR(50),
 city VARCHAR(50),
 join_date DATE
);
```

Orders

```
CREATE TABLE orders (
 order_id INT PRIMARY KEY,
 customer_id INT,
 order_date DATE,
 order_value DECIMAL(10,2)
);
```

Products

```
CREATE TABLE products (
 product_id INT PRIMARY KEY,
 product_name VARCHAR(50),
 category VARCHAR(50),
 price DECIMAL(10,2)
);
```

Order_Items

```
CREATE TABLE order_items (
 order_id INT,
 product_id INT,
 quantity INT
);
```

Employees

```
CREATE TABLE employees (
 emp_id INT PRIMARY KEY,
 emp_name VARCHAR(50),
 department VARCHAR(50),
 salary INT,
 manager_id INT
);
```

Deliveries

```
CREATE TABLE deliveries (
 delivery_id INT PRIMARY KEY,
 order_id INT,
 delivery_time_hours INT,
 status VARCHAR(20)
);
```

Bank_Transactions

```
CREATE TABLE bank_transactions (
 transaction_id INT PRIMARY KEY,
 account_holder VARCHAR(50),
```

```
 transaction_date DATE,
 transaction_type VARCHAR(20),
 amount DECIMAL(10,2)
);
```

SECTION A - BASIC QUERIES

1. Show all customers from Mumbai.
2. Find products costing more than ■5000.
3. Show employees earning between ■55,000 and ■75,000.
4. Find all deliveries marked as delayed.
5. Display customer names in uppercase.

SECTION B - AGGREGATION

1. Find total revenue generated.
2. Find average order value.
3. Count customers city-wise.
4. Find maximum and minimum salary department-wise.
5. Find total quantity sold for each product.

SECTION C - GROUP BY + HAVING

1. Find cities having more than 1 customer.
2. Find departments whose average salary exceeds ■60,000.
3. Show categories generating more than ■25,000 worth of sales.
4. Find account holders having more than 2 transactions.
5. Find customers whose total order value exceeds ■2000.

SECTION D - JOINS

1. Show customer names with their orders.
2. Show product names sold in every order.
3. Show orders along with delivery status.
4. Show employees with their managers.
5. Find total revenue generated by each city.

SECTION E - ADVANCED JOINS

1. Find customers who never placed an order.
2. Find products never sold.
3. Find orders that have no delivery record.
4. Find departments having more employees than Finance.
5. Find cities where customers have placed at least 2 orders.

SECTION F - SUBQUERIES

1. Find employee(s) earning the highest salary.
2. Find customers whose order value is above the overall average.
3. Find products priced above category average.
4. Find the second highest salary.
5. Find customers who placed the maximum number of orders.

SECTION G - WINDOW FUNCTIONS

1. Assign row numbers to employees by salary.
2. Rank employees by salary within each department.
3. Show cumulative transaction amount for each account holder.
4. Show total transactions beside every transaction row.
5. Find the highest order value customer-wise using window functions.

SECTION H - STRING FUNCTIONS

1. Extract first three letters of each customer name.
2. Find length of each product name.
3. Replace 'a' with '@' in customer names.
4. Display product names in lowercase.
5. Concatenate customer name and city.

SECTION I - DATE FUNCTIONS

1. Find number of days since each customer joined.
2. Display month of every order.
3. Find orders placed in February.
4. Find year-wise order count.
5. Find customers who joined before a specific date.

SECTION J - BUSINESS REPORTS

1. City-wise revenue report.
2. Department salary summary report.
3. Delivery performance report.
4. Customer purchase summary report.
5. Bank account transaction summary report.